Assignment -18.1

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Batch no:02

Lab 18 – API Integration: Connecting to External Services with Error Handling

Task 1 – Weather API Integration

Connect to a Weather API to fetch current weather details for a given city.

Instructions:

• Prompt AI to generate Python code using the requests library.

• Handle errors: invalid city name, API key errors, and network failures.

• Display temperature, humidity, and weather description in a readable format.

Expected Output:

• A working Python script that fetches and displays weather data with proper error handling.

Code:

```python
# weather fecth using open meteo api and geocoding api
import requests
def get_coordinates(city):
    url = f"https://geocoding-api.open-meteo.com/v1/search?name={city}"
    try:
        response = requests.get(url, timeout=10)
        response.raise_for_status()
        data = response.json()
        if not data["results"]:
            print("City not found.")
            return None
        return data["results"][0]["latitude"], data["results"][0]["longitude"]
    except requests.exceptions.HTTPError as http_err:
        print(f"HTTP error occurred: {http_err}")
    except requests.exceptions.ConnectionError:
        print("Network error. Please check your connection.")
    except requests.exceptions.Timeout:
        print("Request timed out. Please try again later.")
    except Exception as err:
        print(f"An error occurred: {err}")
    return None
def get_weather(latitude, longitude):
    url = f"https://api.open-meteo.com/v1/forecast?latitude={latitude}&longitude={longitude}&current_weather=true"
    try:
        response = requests.get(url, timeout=10)
        response.raise_for_status()
        data = response.json()
        if "current_weather" not in data:
            print("Weather data not available.")
            return
        temp = data["current_weather"]["temperature"]
        windspeed = data["current_weather"]["windspeed"]
        print(f"Current temperature: {temp}°C")
        print(f"Current wind speed: {windspeed} km/h")
    except requests.exceptions.HTTPError as http_err:
        print(f"HTTP error occurred: {http_err}")
    except requests.exceptions.ConnectionError:
        print("Network error. Please check your connection.")
    except requests.exceptions.Timeout:
        print("Request timed out. Please try again later.")
    except Exception as err:
        print(f"An error occurred: {err}")
if __name__ == "__main__":
    city = input("Enter city name: ")
    coordinates = get_coordinates(city)
```

Output:

```
Current temperature: 34.5°C
Current wind speed: 8.4 km/h
PS C:\Users\shyam\ai assistes code>
Current wind speed: 8.4 km/h
PS C:\Users\shyam\ai assistes code>
```

Task 2 – Currency Exchange Rates

Integrate a Currency Exchange API to convert INR to USD, EUR, and GBP.

Instructions:

- Use AI to generate API request code.
- Validate user input for amount and currency codes.
- Implement error handling for invalid responses and API

downtime.

• Show conversion results clearly in tabular format.

Expected Output:

• A script that converts INR to multiple currencies with graceful

error handling.

Code:

```
lab-23.py
'''
Generate a Python script to convert INR to USD, EUR, and GBP using an exchange rate API.
Display the results in a readable format and handle errors like invalid input or network issues.
'''
import requests
def get_exchange_rate(from_currency, to_currency):
    url = f"https://api.exchangerate-api.com/v4/latest/{from_currency}"
    try:
        response = requests.get(url, timeout=10)
        response.raise_for_status()  # Check for HTTP errors

        data = response.json()
        rates = data.get("rates", {})
        exchange_rate = rates.get(to_currency)

        if exchange_rate is not None:
            print(f"Exchange rate from {from_currency} to {to_currency}: {exchange_rate}")
        else:
            print(f"Currency {to_currency} not found in the exchange rates.")

    except requests.exceptions.HTTPError as http_err:
        print(f"HTTP error occurred: {http_err}")
    except requests.exceptions.ConnectionError:
        print("Network error. Please check your connection.")
    except requests.exceptions.Timeout:
        print("Request timed out. Please try again later.")
    except Exception as err:
        print(f"An error occurred: {err}")
if __name__ == "__main__":
    from_currency = "INR"
    to_currencies = ["USD", "EUR", "GBP"]
    for to_currency in to_currencies:
        get_exchange_rate(from_currency, to_currency)
```

Output:

```
ython.exe' 'c:\Users\shyam\.vscode\extensions\ms-p
shyam\ai assistes code\lab-23.py'
Exchange rate from INR to USD: 0.0106
Exchange rate from INR to EUR: 0.00919
Exchange rate from INR to GBP: 0.00796
PS C:\Users\shyam\ai assistes code>
```

Task 3 – GitHub Repository Info Fetcher

Connect to the GitHub API to fetch details of a repository (e.g., stars,forks, issues).

Instructions:

• Prompt AI to write a Python function to send GET requests to GitHub API.

• Handle rate limit errors, 404 (repo not found), and invalid input.

• Display repository name, description, stars, forks, and open issues.

Expected Output:

• A Python program that retrieves and displays GitHub repository information with robust error handling

code:

lab-23.py

```python
'''
generate a python code on get gate request on github api to fetch user details  Fetch Repository ,
Stars & Forks, Open Issues and handle errors like invalid username, network issues, and timeouts.
'''
import requests

def get_github_user(username):
    url = f"https://api.github.com/users/{username}"
    try:
        response = requests.get(url, timeout=10)
        response.raise_for_status()  # Check for HTTP errors

        data = response.json()
        print(f"User: {data.get('login')}")
        print(f"Name: {data.get('name')}")
        print(f"Bio: {data.get('bio')}")
        print(f"Public repos: {data.get('public_repos')}")

    except requests.exceptions.HTTPError as http_err:
        if http_err.response.status_code == 404:
            print(f"Error: Invalid username '{username}'. User not found.")
        else:
            print(f"HTTP error occurred: {http_err}")
    except requests.exceptions.ConnectionError:
        print("Network error. Please check your connection.")
    except requests.exceptions.Timeout:
        print("Request timed out. Please try again later.")
    except Exception as err:
        print(f"An error occurred: {err}")

if __name__ == "__main__":
    usernames = ["torvalds", "invalid_user_12345", "octocat"]
    for username in usernames:
        get_github_user(username)

```

Output:

```
User: torvalds
Name: Linus Torvalds
Bio: None
Public repos: 11
Error: Invalid username 'invalid_user_12345'. User not found.
User: octocat
Name: The Octocat
Bio: None
User: octocat
Name: The Octocat
Bio: None
Public repos: 8
PS C:\Users\shyam\ai assistes code>
```

Task 4 – Real-Time Application: News Headlines Aggregator

Scenario: Build a News Aggregator using a free News API.

Requirements:

• Fetch top 5 headlines for a given category (sports, technology, health).

• Handle errors like invalid category, missing API key, and request failures.

• Display headlines in a user-friendly numbered list format.

• Include a retry mechanism if the first request fails.

Expected Output:

• A working Python script that retrieves and displays news headlines with proper error handling

code:

```
'''
Generate a Python script using the requests library to build a News Headlines Aggregator.
The program should ask the user to enter a category (sports, technology, or health) and fetch the top 5 headlines from a free news source such as Google News RSS. Display the h
'''
import requests
from xml.etree import ElementTree as ET
import time
def fetch_headlines(category):
    """Fetch top 5 headlines for a given category"""
    # Google News RSS feeds by category
    feeds = {
        'sports': 'https://news.google.com/rss/headlines?gl=US&ceid=US:en&hl=en&topic=SPORTS',
        'technology': 'https://news.google.com/rss/headlines?gl=US&ceid=US:en&hl=en&topic=TECHNOLOGY',
        'health': 'https://news.google.com/rss/headlines?gl=US&ceid=US:en&hl=en&topic=HEALTH'
    }
    if category.lower() not in feeds:
        print(f"Error: Invalid category '{category}'. Available categories: sports, technology, health")
        return None

    url = feeds[category.lower()]
    max_retries = 3
    retry_count = 0
    while retry_count < max_retries:
        try:
            response = requests.get(url, timeout=5)
            response.raise_for_status()
            root = ET.fromstring(response.content)
            headlines = []
            for item in root.findall('.//item')[:5]:
                title = item.find('title')
                if title is not None:
                    headlines.append(title.text)
            return headlines
        except requests.exceptions.RequestException as e:
            retry_count += 1
            if retry_count < max_retries:
                print(f"Request failed. Retrying... ({retry_count}/{max_retries})")
                time.sleep(2)
            else:
                print(f"Error: Failed to fetch headlines after {max_retries} attempts. {str(e)}")
                return None
def main():
    while True:
        category = input("Enter category (sports, technology, or health): ")
        headlines = fetch_headlines(category)
```

```python
            return None
def main():
    while True:
        category = input("Enter category (sports, technology, or health): ")
        headlines = fetch_headlines(category)
        if headlines:
            print(f"\nTop 5 {category.upper()} Headlines:")
            for i, headline in enumerate(headlines, 1):
                print(f"{i}. {headline}")
            break
        else:
            retry = input("Would you like to try again? (yes/no): ")
            if retry.lower() != 'yes':
                break

if __name__ == "__main__":
    main()
```

output:

```
Top 5 HEALTH Headlines:
1. U.S. Circulates Iran Peace Plan While Sending Troops to the Middle East - The New York Times
2. ICE Agents at Some Airports Begin Checking IDs in Security Lines - The New York Times
3. White House holds off on CDC pick as search for permanent chief continues - The Washington Post
4. GOP lawmakers vent frustration over Trump administration’s lack of info on Iran war - CNN
5. TSA boss warns of airport shutdowns, but no deal yet on day 40 of Homeland Security funding fight - AP News
PS C:\Users\shyam\ai assistes code>
```